

VERA X16 for Atari — Running the tests in the emulator

atari800 fork with the VeraX16 PBI card

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Contents

1 Overview	1
2 Prerequisites	1
2.1 Building the emulator	1
2.2 Building the ROM, the drivers and the test disks	1
3 The basic command	2
3.1 Starting a program from DOS 2.0S	2
3.2 Starting a program without a disk	2
4 Options	2
4.1 VeraX16 card options	2
4.2 Useful atari800 options	3
5 Running each test	3
5.1 TESTFX — FX coprocessor	3
5.2 TESTIRQ — VERA interrupts	3
5.3 TESTPLR — VTM player	3
5.4 TEST / TESTGS / TESTMAZ / TESTMTX — driver and screen modes	3
5.5 TESTRMT — RMT player on POKEY + VERA	4
5.6 TESTRIO — music during disk I/O	4
5.7 RUNCPM — ANSI terminal through FujiNet	4
6 Troubleshooting	4

1 Overview

The test programs in **vera-tests/** run on a fork of the **atari800** emulator (version 5.2.0) that emulates the VeraX16 card on the Atari PBI bus: VERA registers at **\$D100-\$D11F**, the 2 KB handler ROM at **\$D800-\$DFFF** (selected through the PBI latch **\$D1FF**).

The emulator lives next to this repository:

```
~/Progetti/ATARI/
├─ atari800/                  emulator (src/atari800 is the binary)
│   └─ vera_pbi_rom -> ../VERA_ATARI_PBI
└─ VERA_ATARI_PBI/          this repository (ROM, drivers, tests)
```

All the commands below are run **from the root of this repository** (VERA_ATARI_PBI), so the emulator is **../atari800/src/atari800**. For brevity:

EMU=../atari800/src/atari800

2 Prerequisites

2.1 Building the emulator

```
cd ../atari800
./autogen.sh
./configure --enable-pbi-verax16
make
cd -
```

The VERA emulation is in `src/pbi_verax16.c` (registers, FX, audio, SPI) and `src/vera_video.c` (rendering). Without `--enable-pbi-verax16` the `-verax16` options do not exist.

No Atari OS ROM is required: when none is configured in `~/atari800.cfg` atari800 uses the built-in AltirraOS and Altirra BASIC.

2.2 Building the ROM, the drivers and the test disks

Required tools: `cc65` (`ca65`, `ld65`, `cl65`), `dir2atr`, `python3`.

```
make # ROM, VERA*.SYS drivers, test programs, disk images
```

Main outputs:

File	Content
<code>vera_pbi_handler.rom</code>	PBI handler ROM, 2 KB (boots in 80×60)
<code>VERA4030.SYS</code> , <code>VERA8030.SYS</code> , <code>VERA8060.SYS</code>	RAM driver for 40×30, 80×30, 80×60
<code>disk1-runcpm.atr</code>	DOS 2.0S, <code>RUNCPM.COM</code> , <code>VERA8030.SYS</code>
<code>disk2-veratests-40x30.atr</code>	<code>TEST4</code> , <code>TESTGS4</code> , <code>TESTMAZ4</code> , <code>TESTMTX4</code> , <code>TESTRMT</code> , <code>VERA4030.SYS</code>
<code>disk2-veratests-80x30.atr</code>	<code>TEST8</code> , <code>TESTGS8</code> , <code>TESTMAZ8</code> , <code>TESTMTX8</code> , <code>TESTRMT</code> , <code>VERA8030.SYS</code>
<code>disk2-veratests-80x60.atr</code>	<code>TEST6</code> , <code>TESTGS6</code> , <code>TESTMAZ6</code> , <code>TESTMTX6</code> , <code>TESTRMT</code> , <code>VERA8060.SYS</code>
<code>disk3-standalone.atr</code>	<code>TESTFX</code> , <code>TESTIRQ</code> , <code>TESTPLR</code> + <code>DEM0.VTM</code> (tests without the driver)
<code>disk4-rmtio.atr</code>	<code>TESTRIO</code> with MyPicoDos autorun and the test files

The `TESTn/TESTGSn/TESTMAZn/TESTMTXn` programs on the `disk2` disks already contain the driver of their resolution ($n = 4$: 40×30, 8: 80×30, 6: 80×60).

3 The basic command

```
$EMU -xl -pal -nopatch -verax16 -verax16-rom vera_pbi_handler.rom \
disk3-standalone.atr
```

- `-verax16` plugs the card in; `-verax16-rom` gives the handler ROM.
- `-xl` (800XL) or `-xe` (130XE): the driver needs an XL/XE machine.
- `-pal` or `-ntsc`: the tests must pass in both.
- `-nopatch`: no high-speed SIO patch, disk I/O runs at the real serial speed (the H: device keeps working). To make it permanent, set `ENABLE_SIO_PATCH=0` in `~/atari800.cfg`.
- The last argument is the disk in `D1`;: more `.atr` files go to `D2`:, `D3`:...

The emulator opens two outputs: the normal ANTIC/GTIA screen and the VERA screen (640×480 VGA).

3.1 Starting a program from DOS 2.0S

The disks boot DOS 2.0S. From the DUP menu:

1. press `L` (BINARY LOAD);
2. type the file name, for example `TESTFX.COM`, and press Return.

3.2 Starting a program without a disk

-run loads an executable straight from the host file system:

```
$EMU -xl -pal -nopatch -verax16 -verax16-rom vera_pbi_handler.rom \  
-run TESTIRQ.COM
```

4 Options

4.1 VeraX16 card options

Option	Meaning
-verax16	enable the card (alias <code>--use-verax16</code>)
-verax16-rom F	handler ROM, 2 KB at \$D800-\$DFFF
-verax16-pbi-id N	PBI device bit 0-7 (default 7, mask \$80)
-verax16-config-ms N	FPGA configuration time after power-on, bus dead meanwhile (default 100); 0 = the board holds Atari RESET until CONFIG_DONE
-verax16-debuglevel N	log level (default 0)
-verax16-psg-volume N	VERA PSG level in percent, 0-400 (default 100 = one voice as loud as a POKEY channel at volume 15)
-verax16-sdcard F	raw SD image (e.g. from <code>dd</code>) exposed through the VERA SPI

4.2 Useful atari800 options

Option	Use
-xl, -xe	800XL / 130XE machine
-pal, -ntsc	video standard
-basic, -nobasic	BASIC ROM on/off
-run F	run a COM/EXE/XEX/BAS file
-nopatch	no high-speed SIO patch: real serial timing, H: still works
-nopatchall	no OS patches: real SIO serial timing (H: does not work)
-volume N	output volume 0-100
-stereo	two POKEYs
-turbo	as fast as possible
-netsio [port]	NetSIO for FujiNet-PC (default UDP 9997)
-config F	alternate configuration file

5 Running each test

5.1 TESTFX — FX coprocessor

```
$EMU -xl -pal -nopatch -verax16 -verax16-rom vera_pbi_handler.rom \  
disk3-standalone.atr
```

From DUP: L, TESTFX.COM. It checks every FX register and measures VRAM copy/fill throughput. Expected result: **PASS 36, FAIL 0**.

5.2 TESTIRQ — VERA interrupts

```
$EMU -xl -pal -nopatch -verax16 -verax16-rom vera_pbi_handler.rom \  
-run TESTIRQ.COM  
$EMU -xl -ntsc -nopatch -verax16 -verax16-rom vera_pbi_handler.rom \  
-run TESTIRQ.COM
```

It checks the VIMIR0 hook: VSYNC at 59.94 Hz, line IRQ, AFLOW masking, chaining to the OS and removal. Expected result: **PASS 13, FAIL 0**, both in PAL and in NTSC. See [Documentation/VERA-IRQ.md](#).

5.3 TESTPLR — VTM player

```
$EMU -xl -pal -nopatch -volume 100 -verax16 -verax16-rom vera_pbi_handler.rom \  
disk3-standalone.atr
```

From DUP: L, TESTPLR.COM. Use `-pal` or `-ntsc` according to the frame rate the song was converted for. `-volume 100` is needed: the VERA PSG is quiet at the default mixer level (or raise `-verax16-psg-volume`).

5.4 TEST / TESTGS / TESTMAZ / TESTMTX — driver and screen modes

```
$EMU -xl -pal -nopatch -verax16 -verax16-rom vera_pbi_handler.rom \  
disk2-veratests-80x30.atr
```

Use the disk of the resolution under test (40x30, 80x30, 80x60) and start, from DUP, TEST8.COM, TESTGS8.COM, TESTMAZ8.COM, TESTMTX8.COM (digit 4 or 6 on the other disks): font loading, gradient, scrolling, maze and matrix demos.

5.5 TESTRMT — RMT player on POKEY + VERA

On every `disk2-veratests-*.atr` disk, it does not need VERA.SYS. From DUP: L, TESTRMT.COM.

Key	Output
1	POKEY only
2	VERA only
3	both (default)
4	hybrid, RMT8 only: POKEY channels 1-4, VERA channels 5-8
S	VERA stereo on/off
ESC	stop and exit

5.6 TESTRIO — music during disk I/O

```
make disk4-rmtio.atr  
$EMU -xl -pal -nopatchall -nobasic \  
-verax16 -verax16-rom vera_pbi_handler.rom disk4-rmtio.atr
```

The disk starts TESTRIO.COM by itself (MyPicoDos autorun). `-nopatchall` is mandatory: without it the emulator intercepts SIO and there is no real serial timing. Expected (PAL, RBL loader): ~720 bytes/s, 0 errors, 0 lost ticks.

5.7 RUNCPM — ANSI terminal through FujiNet

Terminal 1, FujiNet-PC:

```
cd FujiNet/fujinet-pc-ATARI  
./run-fujinet -c fnconfig.ini -s SD/
```

Wait for `### NetSIO stopped ###` in the log, then in terminal 2:

```
$EMU -xl -pal -nopatch -netsio -verax16 -verax16-rom vera_pbi_handler.rom
```

Details in [README-fujinet.md](#).

6 Troubleshooting

Symptom	Cause / fix
ERROR: VeraX16 PBI card not found (\$D100 not responding).	<code>-verax16</code> is missing: every test calls <code>vera_require()</code> at start

Symptom	Cause / fix
Three beeps from the console speaker at boot	the ROM did not see the VERA within ~0.7 s; check <code>-verax16-config-ms</code>
Unknown option <code>-verax16</code>	emulator built without <code>--enable-pbi-verax16</code>
No sound or very quiet VERA	add <code>-volume 100</code> or raise <code>-verax16-psg-volume</code>
A file is on the disk but DOS does not list it, or “0 FREE SECTORS”	<code>dir2atr</code> bug; rebuild with <code>make</code> , which runs <code>vera-tests/tools/fix_atr_vtoc.py</code>
SIO errors or wrong speed in <code>TESTRI0</code>	<code>-nopatchall</code> is missing